Uploaded image preview ### Core Networking Concepts #### 1. Cisco IOS Modes * **User EXEC Mode (`Router>`):** Basic diagnostic mode with limited commands. * **Privileged EXEC Mode (`Router#`):** Accessed using `enable`. Allows viewing detailed system status and running configuration (`show` commands). * **Global Configuration Mode (`Router(config)#`):** Accessed using `configure terminal`. Modifies system-wide settings. * **Interface Configuration Mode (`Router(config-if)#`):** Accessed by specifying an interface (e.g., `interface fa0/0`). Assigns IP addresses, descriptions, and operational states. #### 2. DTE vs. DCE & Clock Rate * **DTE (Data Terminal Equipment):** Router default mode. Receives timing signals from external equipment. * **DCE (Data Communications Equipment):** Synchronizes the WAN link by providing a timing signal. In a lab environment using direct serial cables, the DCE end must be configured with a `clock rate` (e.g., `clock rate 64000` bits per second). * **Clock Icon:** Hovering over a serial link in Packet Tracer displays a small clock icon on the DCE interface. #### 3. Static Routing vs. Default Routing * **Static Route (`ip route [net] [mask] [next-hop]`):** Manually defines a specific path to a remote network. Features low CPU overhead and no routing protocol traffic across WAN links. * **Default Route (`ip route 0.0.0.0 0.0.0.0 [next-hop]`):** Wildcard static route that sends any packet with an unknown destination network to a designated next-hop gateway. Used primarily on stub networks (networks with a single exit path). -- ### Step 1: PC Configuration Configure the IP address, Subnet Mask, and Default Gateway on each end device in Packet Tracer. (Click **PC > Desktop tab > IP Configuration**). #### LAN A (Left Side — Connected to Switch0 / Lab_A Router) Device | IP Address | Subnet Mask | Default Gateway | Connected Router Interface — | — | — | — | — **PC0** | `192.168.10.2` | `255.255.255.0` | `192.168.10.1` | `Lab_A FastEthernet0/0` **PC1** | `192.168.10.3` | `255.255.255.0` | `192.168.10.1` | `Lab_A FastEthernet0/0` **PC4** | `192.168.10.4` | `255.255.255.0` | `192.168.10.1` | `Lab_A FastEthernet0/0` #### LAN B (Right Side — Connected to Switch1 / Lab_B Router) Device | IP Address | Subnet Mask | Default Gateway | Connected Router Interface — | — | — | — | — **PC2** | `192.168.30.2` | `255.255.255.0` | `192.168.30.1` | `Lab_B FastEthernet0/0` **PC3** | `192.168.30.3` | `255.255.255.0` | `192.168.30.1` | `Lab_B FastEthernet0/0` **PC5** | `192.168.30.4` | `255.255.255.0` | `192.168.30.1` | `Lab_B FastEthernet0/0` -- ### Step 2: Basic Router Interface Configuration #### Lab_A Router (Router5 in Diagram) ```text Router> enable Router# configure terminal Router(config)# hostname Lab_A ! Configure LAN interface (Gateway for PC0, PC1, PC4) Lab_A(config)# interface FastEthernet0/0 Lab_A(config-if)# ip address 192.168.10.1 255.255.255.0 Lab_A(config-if)# description Lab_A LAN Connection Lab_A(config-if)# no shutdown Lab_A(config-if)# exit ! Configure WAN interface to Lab_B Lab_A(config)# interface Serial1/0

Lab_A(config-if)# ip address 192.168.20.1 255.255.255.0 Lab_A(config-if)# description WAN Connection to Lab_B Lab_A(config-if)# no shutdown Lab_A(config-if)# exit ! Configure secondary serial interface Lab_A(config)# interface Serial1/1 Lab_A(config-if)# ip address 192.168.50.1 255.255.255.0 Lab_A(config-if)# no shutdown Lab_A(config-if)# exit ! Configure MOTD Banner and Save Settings Lab_A(config)# banner motd # This is the Lab_A router # Lab_A(config)# end Lab_A# copy running-config startup-config ``` #### Lab_B Router (Router4 in Diagram) ```text Router> enable Router# configure terminal Router(config)# hostname Lab_B ! Configure LAN interface (Gateway for PC2, PC3, PC5) Lab_B(config)# interface FastEthernet0/0 Lab_B(config-if)# ip address 192.168.30.1 255.255.255.0 Lab_B(config-if)# description Lab_B LAN Connection Lab_B(config-if)# no shutdown Lab_B(config-if)# exit ! Configure WAN interface (DCE side requires clock rate) Lab_B(config)# interface Serial1/0 Lab_B(config-if)# ip address 192.168.20.2 255.255.255.0 Lab_B(config-if)# description WAN Connection to Lab_A Lab_B(config-if)# clock rate 64000 Lab_B(config-if)# no shutdown Lab_B(config-if)# exit ! Configure secondary serial interface Lab_B(config)# interface Serial1/1 Lab_B(config-if)# ip address 192.168.40.1 255.255.255.0 Lab_B(config-if)# no shutdown Lab_B(config-if)# exit ! Configure MOTD Banner and Save Settings Lab_B(config)# banner motd # This is the Lab_B router # Lab_B(config)# end Lab_B# copy running-config startup-config ``` -– ### Step 3: Configuring Static Routing Without routing rules, Lab_A only knows networks `192.168.10.0`, `192.168.20.0`, and `192.168.50.0`. Lab_B only knows `192.168.30.0`, `192.168.20.0`, and `192.168.40.0`. #### Configure Static Routes on Lab_A ```text Lab_A# configure terminal ! Route to LAN B network via Lab_B's serial interface IP Lab_A(config)# ip route 192.168.30.0 255.255.255.0 192.168.20.2 ! Route to Lab_B's secondary network Lab_A(config)# ip route 192.168.40.0 255.255.255.0 192.168.20.2 Lab_A(config)# end Lab_A# copy running-config startup-config ``` #### Configure Static Routes on Lab_B ```text Lab_B# configure terminal ! Route to LAN A network via Lab_A's serial interface IP Lab_B(config)# ip route 192.168.10.0 255.255.255.0 192.168.20.1 ! Route to Lab_A's secondary network Lab_B(config)# ip route 192.168.50.0 255.255.255.0 192.168.20.1 Lab_B(config)# end Lab_B# copy running-config startup-config ``` #### Verification Commands for Static Routing Execute the following commands in Privileged EXEC Mode (`#`) to confirm routing configuration: ```text Lab_A# show ip route ``` **Expected Routing Table Output for Lab_A:** ```text C 192.168.10.0/24 is directly connected, FastEthernet0/0 C 192.168.20.0/24 is directly connected, Serial1/0 S 192.168.30.0/24 [1/0] via 192.168.20.2 S 192.168.40.0/24 [1/0] via 192.168.20.2 C 192.168.50.0/24 is directly connected, Serial1/1 ``` *(Legend: `C` = Directly Connected, `S` = Static Route)* -– ### Step 4:

Configuring Default Routing Default routing simplifies routing table entries by using `0.0.0.0 0.0.0.0` as a catchment for any network destination. #### 1. Remove Existing Static Routes & Apply Default Route on Lab_A ```text Lab_A# configure terminal Lab_A(config)# no ip route 192.168.30.0 255.255.255.0 192.168.20.2 Lab_A(config)# no ip route 192.168.40.0 255.255.255.0 192.168.20.2 Lab_A(config)# ip route 0.0.0.0 0.0.0.0 192.168.20.2 Lab_A(config)# end ``` #### 2. Remove Existing Static Routes & Apply Default Route on Lab_B ```text Lab_B# configure terminal Lab_B(config)# no ip route 192.168.10.0 255.255.255.0 192.168.20.1 Lab_B(config)# no ip route 192.168.50.0 255.255.255.0 192.168.20.1 Lab_B(config)# ip route 0.0.0.0 0.0.0.0 192.168.20.1 Lab_B(config)# end ``` #### Verification Commands for Default Routing ```text Lab_A# show ip route ``` **Expected Routing Table Output for Lab_A:** ```text Gateway of last resort is 192.168.20.2 to network 0.0.0.0 C 192.168.10.0/24 is directly connected, FastEthernet0/0 C 192.168.20.0/24 is directly connected, Serial1/0 C 192.168.50.0/24 is directly connected, Serial1/1 S* 0.0.0.0/0 [1/0] via 192.168.20.2 ``` *(Legend: `S*` = Candidate Default Route)* -– ### Step 5: End-to-End Connectivity Testing Open the **Command Prompt** on **PC0** (`192.168.10.2`) and run pings across the networks: 1. **Test local default gateway:** `ping 192.168.10.1` *(Should reply)* 2. **Test remote WAN link interface:** `ping 192.168.20.2` *(Should reply)* 3. **Test remote PC in LAN B (PC2):** `ping 192.168.30.2` *(First packet may time out due to ARP resolution; subsequent packets will succeed)* i dont udnerstand a sngle things just make a full correct soution correct ips just take tehse sequences of pcs also most diffcult take i take was ips confuion with difeerrnt to everyone please also clear my ips concept please also ok### Core Networking Concepts #### 1. Cisco IOS Modes * **User EXEC Mode (`Router>`):** Basic diagnostic mode with limited commands. * **Privileged EXEC Mode (`Router#`):** Accessed using `enable`. Allows viewing detailed system status and running configuration (`show` commands). * **Global Configuration Mode (`Router(config)#`):** Accessed using `configure terminal`. Modifies system-wide settings. * **Interface Configuration Mode (`Router(config-if)#`):** Accessed by specifying an interface (e.g., `interface fa0/0`). Assigns IP addresses, descriptions, and operational states. #### 2. DTE vs. DCE & Clock Rate * **DTE (Data Terminal Equipment):** Router default mode. Receives timing signals from external equipment. * **DCE (Data Communications Equipment):** Synchronizes the WAN link by providing a timing signal. In a lab environment using direct serial cables, the DCE end must be configured with a `clock rate` (e.g., `clock rate 64000` bits per second). * **Clock Icon:** Hovering over a serial link in Packet Tracer displays a small clock icon on the DCE interface. #### 3. Static Routing vs. Default Routing * **Static Route (`ip route [net] [mask] [next-hop]`):** Manually defines a specific path to a remote network. Features low CPU overhead

and no routing protocol traffic across WAN links. * **Default Route (`ip route 0.0.0.0 0.0.0.0 [next-hop]`):** Wildcard static route that sends any packet with an unknown destination network to a designated next-hop gateway. Used primarily on stub networks (networks with a single exit path). — ### Step 1: PC Configuration Configure the IP address, Subnet Mask, and Default Gateway on each end device in Packet Tracer. (Click **PC > Desktop tab > IP Configuration**). #### LAN A (Left Side — Connected to Switch0 / Lab_A Router) Device | IP Address | Subnet Mask | Default Gateway | Connected Router Interface — | — | — | — | — **PC0** | `192.168.10.2` | `255.255.255.0` | `192.168.10.1` | `Lab_A FastEthernet0/0` **PC1** | `192.168.10.3` | `255.255.255.0` | `192.168.10.1` | `Lab_A FastEthernet0/0` **PC4** | `192.168.10.4` | `255.255.255.0` | `192.168.10.1` | `Lab_A FastEthernet0/0` #### LAN B (Right Side — Connected to Switch1 / Lab_B Router) Device | IP Address | Subnet Mask | Default Gateway | Connected Router Interface — | — | — | — | — **PC2** | `192.168.30.2` | `255.255.255.0` | `192.168.30.1` | `Lab_B FastEthernet0/0` **PC3** | `192.168.30.3` | `255.255.255.0` | `192.168.30.1` | `Lab_B FastEthernet0/0` **PC5** | `192.168.30.4` | `255.255.255.0` | `192.168.30.1` | `Lab_B FastEthernet0/0` — ### Step 2: Basic Router Interface Configuration #### Lab_A Router (Router5 in Diagram) ```text Router> enable Router# configure terminal Router(config)# hostname Lab_A ! Configure LAN interface (Gateway for PC0, PC1, PC4) Lab_A(config)# interface FastEthernet0/0 Lab_A(config-if)# ip address 192.168.10.1 255.255.255.0 Lab_A(config-if)# description Lab_A LAN Connection Lab_A(config-if)# no shutdown Lab_A(config-if)# exit ! Configure WAN interface to Lab_B Lab_A(config)# interface Serial1/0 Lab_A(config-if)# ip address 192.168.20.1 255.255.255.0 Lab_A(config-if)# description WAN Connection to Lab_B Lab_A(config-if)# no shutdown Lab_A(config-if)# exit ! Configure secondary serial interface Lab_A(config)# interface Serial1/1 Lab_A(config-if)# ip address 192.168.50.1 255.255.255.0 Lab_A(config-if)# no shutdown Lab_A(config-if)# exit ! Configure MOTD Banner and Save Settings Lab_A(config)# banner motd # This is the Lab_A router # Lab_A(config)# end Lab_A# copy running-config startup-config ``` #### Lab_B Router (Router4 in Diagram) ```text Router> enable Router# configure terminal Router(config)# hostname Lab_B ! Configure LAN interface (Gateway for PC2, PC3, PC5) Lab_B(config)# interface FastEthernet0/0 Lab_B(config-if)# ip address 192.168.30.1 255.255.255.0 Lab_B(config-if)# description Lab_B LAN Connection Lab_B(config-if)# no shutdown Lab_B(config-if)# exit ! Configure WAN interface (DCE side requires clock rate) Lab_B(config)# interface Serial1/0 Lab_B(config-if)# ip address 192.168.20.2 255.255.255.0 Lab_B(config-if)# description WAN Connection to Lab_A Lab_B(config-if)# clock rate 64000 Lab_B(config-if)# no shutdown Lab_B(config-if)# exit ! Configure secondary serial interface

Lab_B(config)# interface Serial1/1 Lab_B(config-if)# ip address 192.168.40.1 255.255.255.0 Lab_B(config-if)# no shutdown Lab_B(config-if)# exit ! Configure MOTD Banner and Save Settings Lab_B(config)# banner motd # This is the Lab_B router # Lab_B(config)# end Lab_B# copy running-config startup-config ``` –– ### Step 3: Configuring Static Routing Without routing rules, Lab_A only knows networks `192.168.10.0`, `192.168.20.0`, and `192.168.50.0`. Lab_B only knows `192.168.30.0`, `192.168.20.0`, and `192.168.40.0`. #### Configure Static Routes on Lab_A ```text Lab_A# configure terminal ! Route to LAN B network via Lab_B's serial interface IP Lab_A(config)# ip route 192.168.30.0 255.255.255.0 192.168.20.2 ! Route to Lab_B's secondary network Lab_A(config)# ip route 192.168.40.0 255.255.255.0 192.168.20.2 Lab_A(config)# end Lab_A# copy running-config startup-config ``` #### Configure Static Routes on Lab_B ```text Lab_B# configure terminal ! Route to LAN A network via Lab_A's serial interface IP Lab_B(config)# ip route 192.168.10.0 255.255.255.0 192.168.20.1 ! Route to Lab_A's secondary network Lab_B(config)# ip route 192.168.50.0 255.255.255.0 192.168.20.1 Lab_B(config)# end Lab_B# copy running-config startup-config ``` #### Verification Commands for Static Routing Execute the following commands in Privileged EXEC Mode (`#`) to confirm routing configuration: ```text Lab_A# show ip route ``` **Expected Routing Table Output for Lab_A:** ```text C 192.168.10.0/24 is directly connected, FastEthernet0/0 C 192.168.20.0/24 is directly connected, Serial1/0 S 192.168.30.0/24 [1/0] via 192.168.20.2 S 192.168.40.0/24 [1/0] via 192.168.20.2 C 192.168.50.0/24 is directly connected, Serial1/1 ``` *(Legend: `C` = Directly Connected, `S` = Static Route)* –– ### Step 4: Configuring Default Routing Default routing simplifies routing table entries by using `0.0.0.0 0.0.0.0` as a catchment for any network destination. #### 1. Remove Existing Static Routes & Apply Default Route on Lab_A ```text Lab_A# configure terminal Lab_A(config)# no ip route 192.168.30.0 255.255.255.0 192.168.20.2 Lab_A(config)# no ip route 192.168.40.0 255.255.255.0 192.168.20.2 Lab_A(config)# ip route 0.0.0.0 0.0.0.0 192.168.20.2 Lab_A(config)# end ``` #### 2. Remove Existing Static Routes & Apply Default Route on Lab_B ```text Lab_B# configure terminal Lab_B(config)# no ip route 192.168.10.0 255.255.255.0 192.168.20.1 Lab_B(config)# no ip route 192.168.50.0 255.255.255.0 192.168.20.1 Lab_B(config)# ip route 0.0.0.0 0.0.0.0 192.168.20.1 Lab_B(config)# end ``` #### Verification Commands for Default Routing ```text Lab_A# show ip route ``` **Expected Routing Table Output for Lab_A:** ```text Gateway of last resort is 192.168.20.2 to network 0.0.0.0 C 192.168.10.0/24 is directly connected, FastEthernet0/0 C 192.168.20.0/24 is directly connected, Serial1/0 C 192.168.50.0/24 is directly connected, Serial1/1 S* 0.0.0.0/0 [1/0] via 192.168.20.2 ``` *(Legend: `S*` = Candidate Default Route)* –– ### Step 5: End-to-End Connectivity Testing Open the **Command Prompt** on

PC0 (`192.168.10.2`) and run pings across the networks: 1. **Test local default gateway:** `ping 192.168.10.1` *(Should reply)* 2. **Test remote WAN link interface:** `ping 192.168.20.2` *(Should reply)* 3. **Test remote PC in LAN B (PC2):** `ping 192.168.30.2` *(First packet may time out due to ARP resolution; subsequent packets will succeed)* i dont udnerstand a sngle things just make a full correct soution correct ips just take tehse sequences of pcs also most diffcult take i take was ips confuion with difeerrnt to everyone please also clear my ips concept please also okexplain evry term use in this lab manual data like an defin this wth examples so i can also prepare for my laba vivii just have serila0/0 serial0/1 for both laba and lab b router a full final solution correct cmds ok??

Part 1: IP Concepts & Viva Study Guide

1. Real-World Analogy: Understanding IP Addressing

Think of your network like a city:

- **IP Address (192.168.10.2):** Your complete home address. It consists of two parts:
    - **Network ID (192.168.10):** The **Street Name**. All devices on the same local network must share the exact same street name.
    - **Host ID (.2):** Your **House Number**. Every device on the same street must have a unique house number.
- **Subnet Mask (255.255.255.0):** The dividing line. 255 tells the router: *"The first 3 numbers are the street name."* 0 tells the router: *"The last number is the house number."*
- **Default Gateway (192.168.10.1):** The **Exit Gate / Main Guard Post** of your gated community.
    - If PC0 (192.168.10.2) wants to talk to PC1 (192.168.10.3), they are on the **same street** (192.168.10), so they send messages directly without asking the gateway.
    - If PC0 (192.168.10.2) wants to talk to PC3 (192.168.30.3), they are on a **different street** (192.168.30). PC0 cannot deliver it directly, so it sends the packet to its **Default Gateway** (192.168.10.1 on Router LabA) to forward it out of the network.

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2. Key Terminology for Lab Viva

- **Cisco IOS EXEC Modes:**

1. **User EXEC (Router>):** Read-only mode for basic checks.
2. **Privileged EXEC (Router#):** Entered using `enable`. Allows viewing detailed configurations (`show ip route`, `show ip interface brief`).
3. **Global Configuration (Router(config)#):** Entered using `configure terminal`. Used to change router settings (e.g., hostname, static routes).
4. **Interface Configuration (Router(config-if)#):** Entered by selecting a specific port (e.g., `interface serial 0/0`). Used to set IP addresses and turn ports on.

- **DTE vs. DCE & Clock Rate:**
  - **DTE (Data Terminal Equipment):** The router acting as a standard client device receiving signals.
  - **DCE (Data Communications Equipment):** The device that controls timing/speed on a WAN serial link. In real life, ISP equipment does this. In a lab, one router port must act as DCE and set the speed using `clock rate 64000` (64 kbps).
- **Static Routing (ip route [dest_network] [mask] [next_hop]):**
  - Manually instructing a router where to forward packets destined for non-directly connected networks.
  - **Next-Hop Address:** The IP address of the **neighboring router's receiving port** (192.168.20.2).
  - **Exit Interface:** The local router's outgoing port (Serial0/0).
- **Default Routing (ip route 0.0.0.0 0.0.0.0 [next-hop]):**
  - A wildcard route that says: *"If a destination network is not in your routing table, send it to this gateway."* Used on **stub networks** (networks with only one path out).
- **Administrative Distance (AD):**
  - A rating of route trustworthiness (lower is better). Directly connected networks have AD 0, static routes have AD 1, and dynamic routing protocols (like OSPF) have higher ADs.
- **Banner MOTD (Message of the Day):**
  - A legal/warning banner displayed when a user connects to the router CLI.

Part 2: Complete Lab Solution

Based on your actual Packet Tracer setup (image_6de6e1.png), the left side contains **PC0, PC1, PC2**, and the right side contains **PC3, PC4, PC5**. Your router ports are

mapped to **Serial0/0** (WAN Link between routers) and **Serial0/1** (secondary WAN port).

Step 1: End-Device (PC) Configuration

Click each **PC** > **Desktop** > **IP Configuration** and enter the following settings:

Left Side — LAN A (Connected to Switch0 / LabA)

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

Right Side — LAN B (Connected to Switch1 / LabB)

Device	IP Address	Subnet Mask	Default Gateway
PC3	192.168.30.2	255.255.255.0	192.168.30.1
PC4	192.168.30.3	255.255.255.0	192.168.30.1
PC5	192.168.30.4	255.255.255.0	192.168.30.1

Step 2: Basic Interface Configuration

1. Router LabA Configuration

Click **LabA** > **CLI** tab:

```
Router> enable
Router# configure terminal
Router(config)# hostname LabA

! Configure LAN Interface (Gateway for PC0, PC1, PC2)
LabA(config)# interface FastEthernet0/0
LabA(config-if)# ip address 192.168.10.1 255.255.255.0
LabA(config-if)# description LabA LAN Connection
LabA(config-if)# no shutdown
LabA(config-if)# exit

! Configure WAN Interface to LabB (Using Serial0/0)
```

```
LabA(config)# interface Serial0/0
LabA(config-if)# ip address 192.168.20.1 255.255.255.0
LabA(config-if)# description WAN Connection to LabB
LabA(config-if)# no shutdown
LabA(config-if)# exit

! Configure Secondary WAN Interface
LabA(config)# interface Serial0/1
LabA(config-if)# ip address 192.168.50.1 255.255.255.0
LabA(config-if)# no shutdown
LabA(config-if)# exit

! Set MOTD Banner and Save
LabA(config)# banner motd # This is the Lab_A router #
LabA(config)# end
LabA# copy running-config startup-config
```

2. Router LabB Configuration

Click **LabB** > **CLI** tab:

```
Router> enable
Router# configure terminal
Router(config)# hostname LabB

! Configure LAN Interface (Gateway for PC3, PC4, PC5)
LabB(config)# interface FastEthernet0/0
LabB(config-if)# ip address 192.168.30.1 255.255.255.0
LabB(config-if)# description LabB LAN Connection
LabB(config-if)# no shutdown
LabB(config-if)# exit

! Configure WAN Interface to LabA (DCE Side requires clock rate)
LabB(config)# interface Serial0/0
LabB(config-if)# ip address 192.168.20.2 255.255.255.0
LabB(config-if)# description WAN Connection to LabA
LabB(config-if)# clock rate 64000
LabB(config-if)# no shutdown
LabB(config-if)# exit

! Configure Secondary WAN Interface
LabB(config)# interface Serial0/1
LabB(config-if)# ip address 192.168.40.1 255.255.255.0
```

```
LabB(config-if)# no shutdown
LabB(config-if)# exit

! Set MOTD Banner and Save
LabB(config)# banner motd # This is the Lab_B router #
LabB(config)# end
LabB# copy running-config startup-config
```

Step 3: Static Routing Configuration

Tell **LabA** how to find networks 192.168.30.0 and 192.168.40.0, and tell **LabB** how to find networks 192.168.10.0 and 192.168.50.0.

On LabA Router:

```
LabA# configure terminal
LabA(config)# ip route 192.168.30.0 255.255.255.0 192.168.20.2
LabA(config)# ip route 192.168.40.0 255.255.255.0 192.168.20.2
LabA(config)# end
LabA# copy running-config startup-config
```

On LabB Router:

```
LabB# configure terminal
LabB(config)# ip route 192.168.10.0 255.255.255.0 192.168.20.1
LabB(config)# ip route 192.168.50.0 255.255.255.0 192.168.20.1
LabB(config)# end
LabB# copy running-config startup-config
```

Verify Static Routing on LabA:

```
LabA# show ip route
```

Expected Routing Output:

```
C    192.168.10.0/24 is directly connected, FastEthernet0/0
C    192.168.20.0/24 is directly connected, Serial0/0
S    192.168.30.0/24 [1/0] via 192.168.20.2
S    192.168.40.0/24 [1/0] via 192.168.20.2
C    192.168.50.0/24 is directly connected, Serial0/1
```

Step 4: Default Routing Configuration

To replace individual static routes with a single wildcard default route (0.0.0.0 0.0.0.0):

On LabA Router:

```
LabA# configure terminal
! Remove individual static routes
LabA(config)# no ip route 192.168.30.0 255.255.255.0 192.168.20.2
LabA(config)# no ip route 192.168.40.0 255.255.255.0 192.168.20.2

! Add default route
LabA(config)# ip route 0.0.0.0 0.0.0.0 192.168.20.2
LabA(config)# end
LabA# copy running-config startup-config
```

On LabB Router:

```
LabB# configure terminal
! Remove individual static routes
LabB(config)# no ip route 192.168.10.0 255.255.255.0 192.168.20.1
LabB(config)# no ip route 192.168.50.0 255.255.255.0 192.168.20.1

! Add default route
LabB(config)# ip route 0.0.0.0 0.0.0.0 192.168.20.1
LabB(config)# end
LabB# copy running-config startup-config
```

Verify Default Routing on LabA:

```
LabA# show ip route
```

Expected Routing Output:

```
Gateway of last resort is 192.168.20.2 to network 0.0.0.0

C    192.168.10.0/24 is directly connected, FastEthernet0/0
C    192.168.20.0/24 is directly connected, Serial0/0
C    192.168.50.0/24 is directly connected, Serial0/1
S*   0.0.0.0/0 [1/0] via 192.168.20.2
```

Step 5: Connectivity Verification (Ping Test)

1. Open **PC0** > **Desktop** > **Command Prompt**.
2. Run the following commands:

- **Ping Local Gateway:** ping 192.168.10.1 *(Tests PC to local router link)*
- **Ping Remote Router WAN Port:** ping 192.168.20.2 *(Tests router-to-router WAN link)*
- **Ping Remote PC across WAN:** ping 192.168.30.2 *(Tests full end-to-end routing)*

(Note: The very first ping to 192.168.30.2 may show Request timed out once while ARP resolves. Ping a second time to see 4 successful replies).